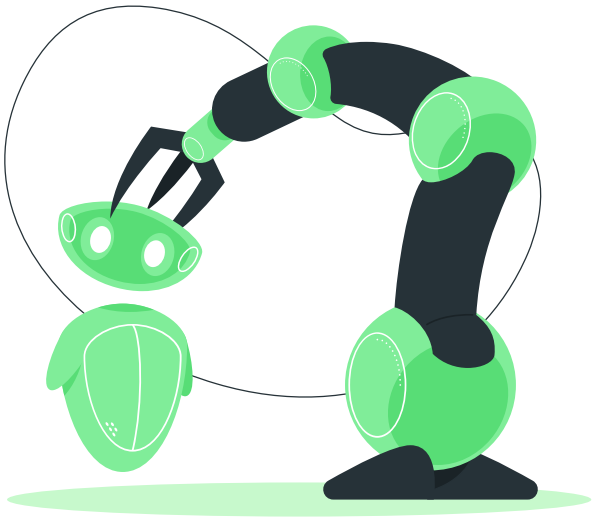
Robotics
CS 3015

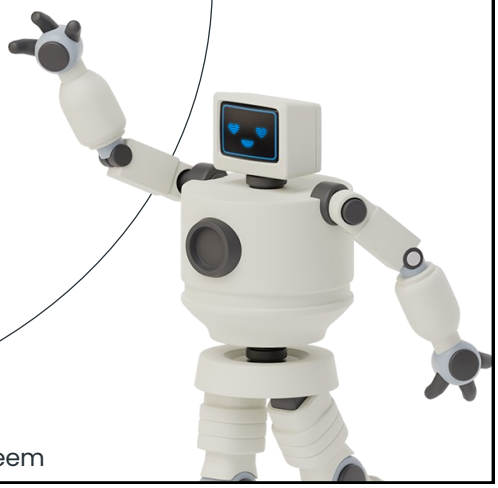
Umarah Qaseem



1

5

- DH Notation and Activity
- Workspace
- Assignment 1
- Link Transformations
- Robot Arm Kinematics
- Forward Kinematics
- Transforms



2

Degree of Freedom?



4

Denavit–Hartenberg notation



5

Denavit-Hartenberg notation

- Developed a general theory to describe an articulated sequence of joints
- Only applicable to serial link mechanisms NOT parallel mechanisms

6

Each joint in the robot is described by four parameters

7

Denavit-Hartenberg notation

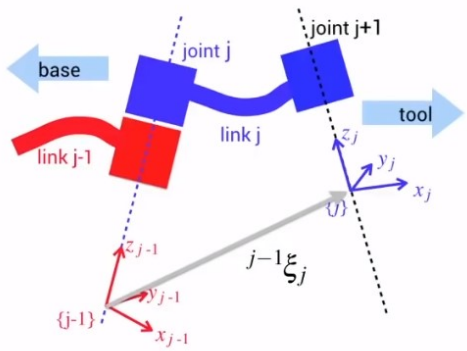
Joint angle	θ_j	the angle between the x_{j-1} and x_j axes about the z_{j-1} axis	revolute joint variable
Link offset	d_j	the distance from the origin of frame $j-1$ to the x_j axis along the z_{j-1} axis	prismatic joint variable
Link length	a_j	the distance between the z_{j-1} and z_j axes along the x_j axis; for intersecting axes is parallel to $\hat{z}_{j-1} \times \hat{z}_j$	constant
Link twist	α_j	the angle from the z_{j-1} axis to the z_j axis about the x_j axis	constant
Joint type	σ_j	$\sigma = R$ for a revolute joint, $\sigma = P$ for a prismatic joint	constant

$$^{j-1}\xi_j(\theta_j, d_j, a_j, \alpha_j) = \mathcal{R}_z(\theta_j) \oplus \mathcal{T}_z(d_j) \oplus \mathcal{T}_x(a_j) \oplus \mathcal{R}_x(\alpha_j)$$

DH parameters

$$^{j-1}\xi_j \sim \mathbf{A} = \mathbf{R}_z(\theta_j)\mathbf{T}_z(d_j)\mathbf{T}_x(a_j)\mathbf{R}_x(\alpha_j)$$

- Only 4 parameters
- *θ, d, a, α*



Scientists – Denavit and Hartenberg



Jacques Denavit and Richard Hartenberg

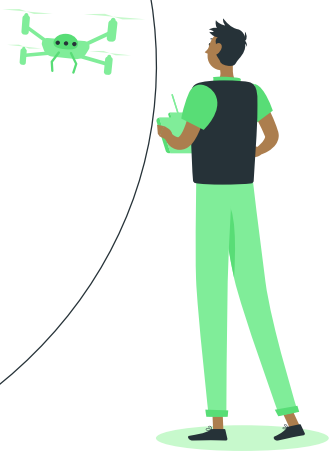
10

DH notation conventions

- There are two conventions for DH parameters
  - Standard
  - Modified
- Always mention which convention you have used whenever you publish any research or write/use DH parameters anywhere.

11

One more look & Examples



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12

DH pa



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13

DH parameters

- 4 parameters *θ, d, a, α*
- The two variables used for rotation are θ and α.
- The two variables used for displacement are a (sometimes also referred as r) and d.

14

Objective

- Find the pose between frame n-1 and frame n

$$^{j-1}\xi_j \sim \mathbf{A} = \mathbf{R}_z(\theta_j)\mathbf{T}_z(d_j)\mathbf{T}_x(a_j)\mathbf{R}_x(\alpha_j)$$

15

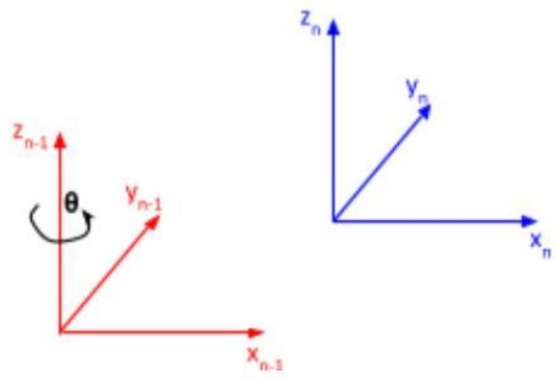
Difference

- The difference between standard and modified notation is…?

16

Theta

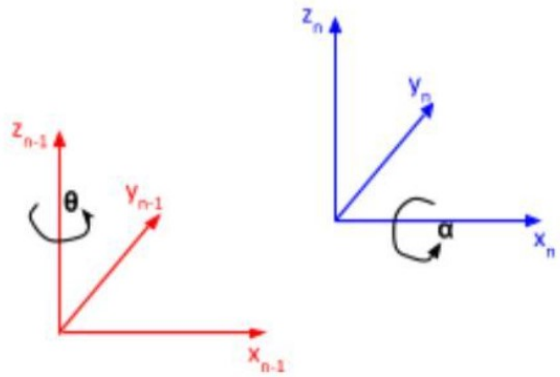
- θ is the angle from x_{n-1} to x_n around z_{n-1}.



17

Alpha

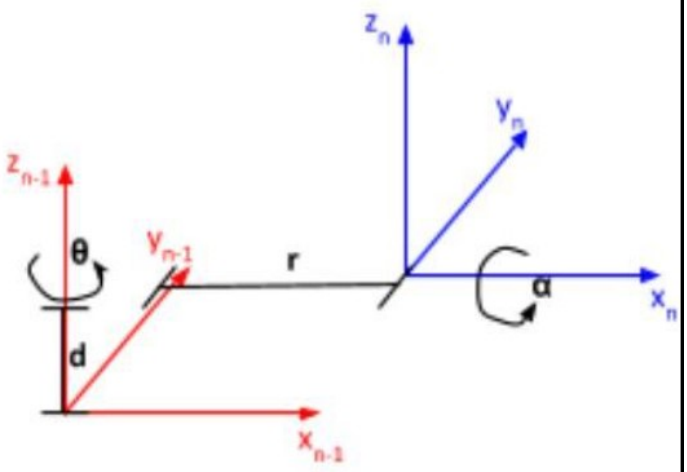
- α is the angle from z_{n-1} to z_n around x_n.



18

Distance d

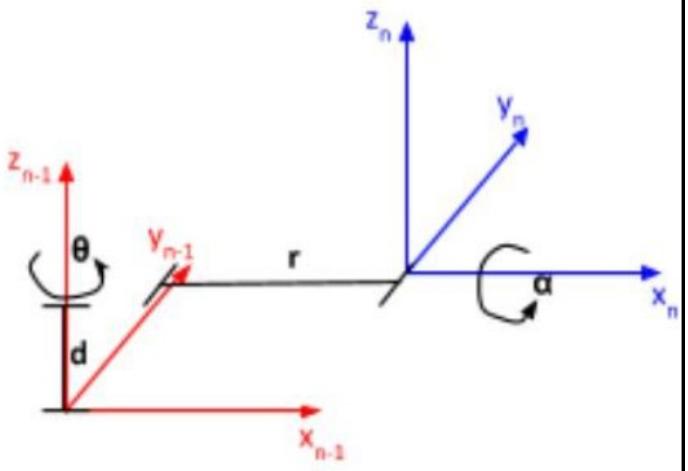
- d is the distance from x_{n-1} to x_n alon the z_{n-1} direction.



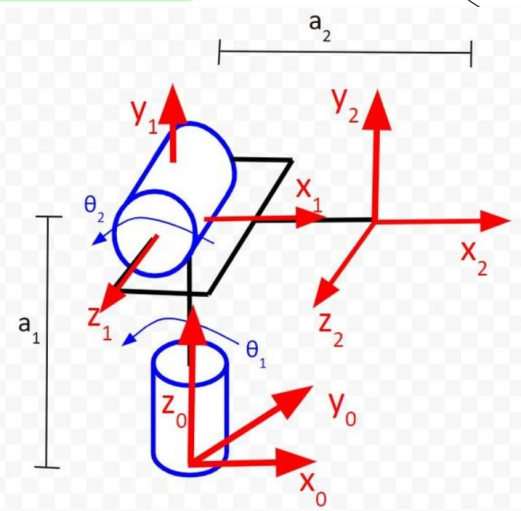
19

Distance a or r

- a (or r) is the distance between the origin of the n-1 frame and the origin of the n frame along the x_n direction.

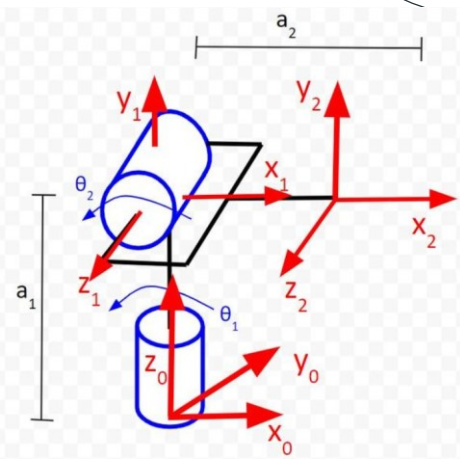


Example 1 – Two Degree of Freedom Robotic Arm

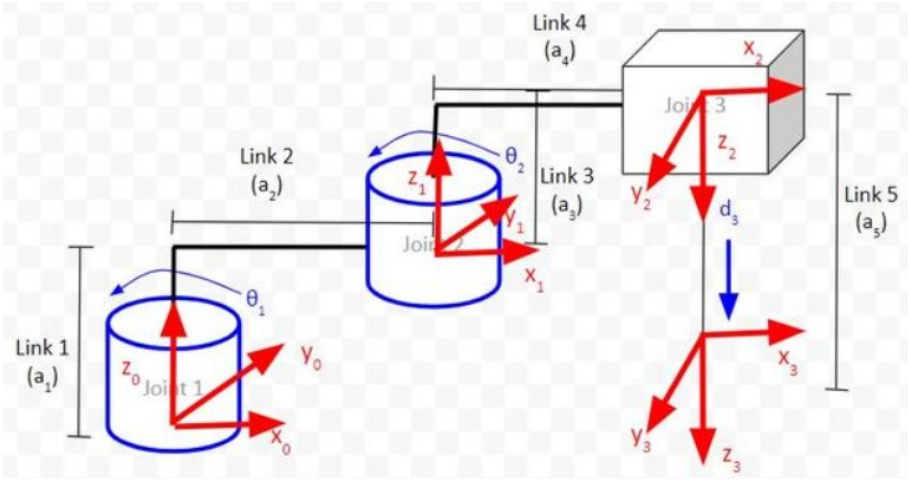


Example 1 – Two Degree of Freedom Robotic Arm

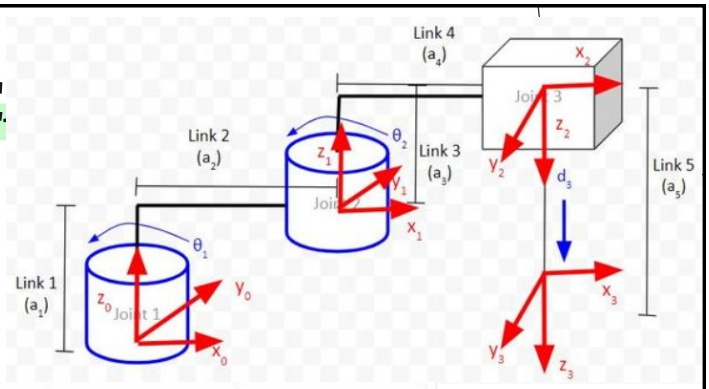
Joint i	θ_i (deg)	α_i (deg)	r_i (cm)	d_i (cm)
1	θ_1	90	0	a_1
2	θ_2	0	a_2	0



Example 2 – SCARA Robot



Example 2 – SC



Joint i	θ_i (deg)	α_i (deg)	r_i (cm)	d_i (cm)
1	θ_1	0	a_2	a_1
2	θ_2	180	a_4	a_3
3	0	0	0	$a_5 + d_3$

24

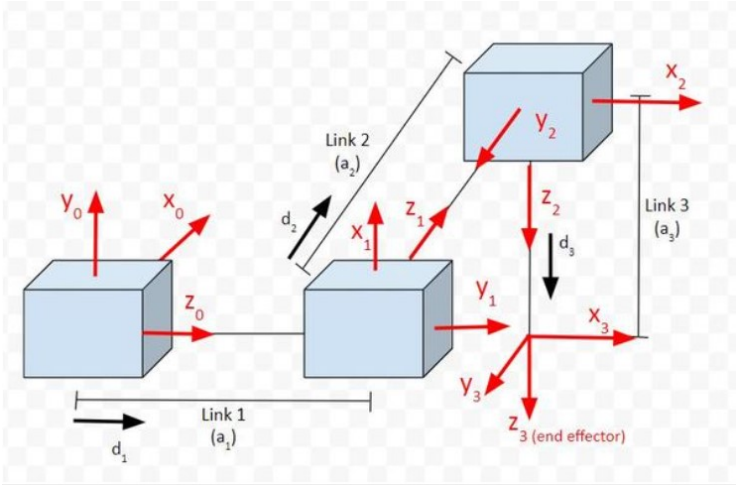
Example 3 – Cartesian Robot



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25

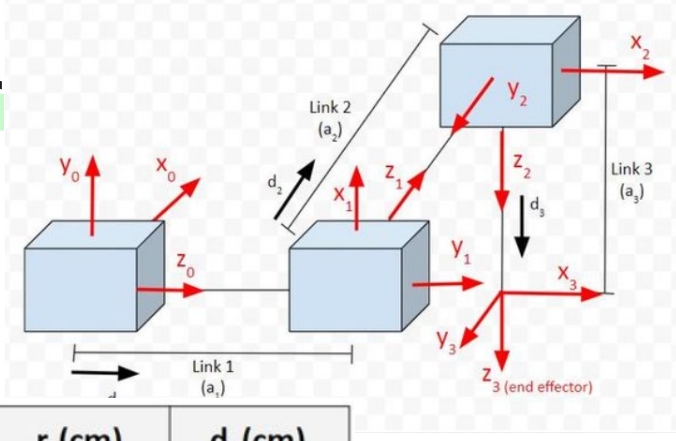
Example 3 – Cartesian Robot



26

If Theta is shown as a variable then, it means the joint is revolutory otherwise joint is prisimitive.

Example 3 – Car



Joint i	θ_i (deg)	α_i (deg)	r_i (cm)	d_i (cm)
1	90	90	0	$a_1 + d_1$
2	90	-90	0	$a_2 + d_2$
3	0	0	0	$a_3 + d_3$

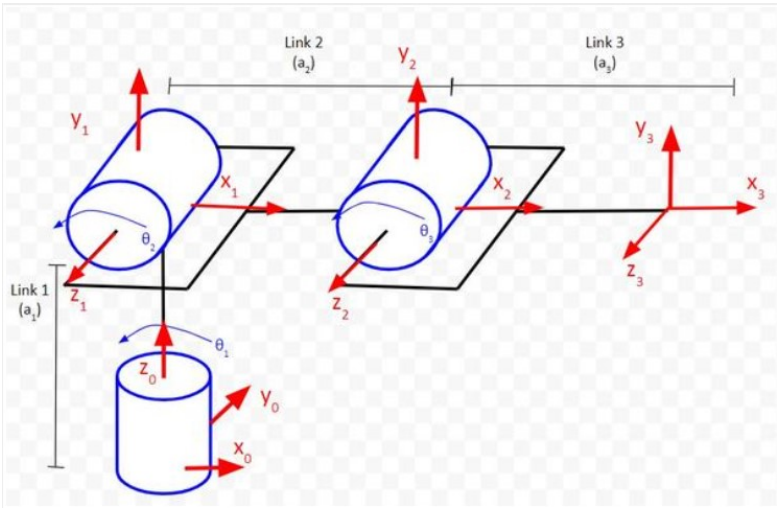
27

Example 4 – Articulated Robot



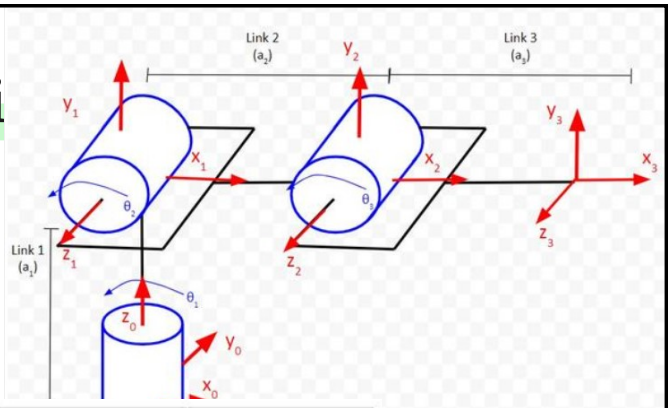
28

Example 4 – Articulated Robot



29

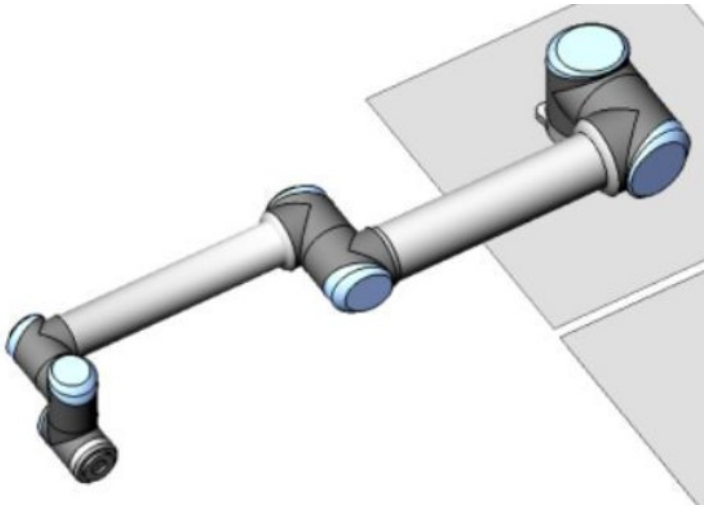
Example 4 – Arti



Joint i	θ_i (deg)	α_i (deg)	r_i (cm)	d_i (cm)
1	θ_1	90	0	a_1
2	θ_2	0	a_2	0
3	θ_3	0	a_3	0

30

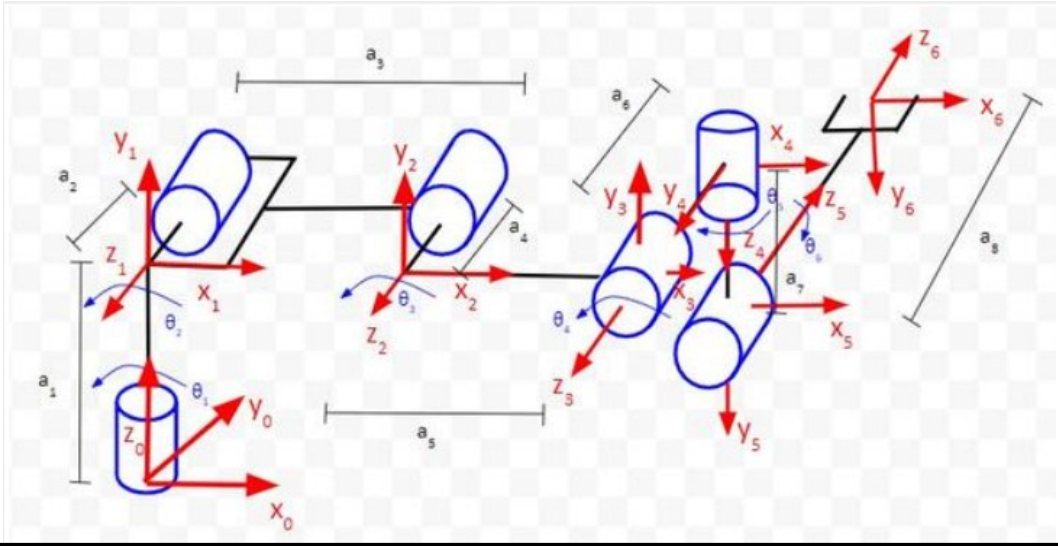
Example 6 – Six DOF Collaborative Robot



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34

Example 6 – Six DOF Collaborative Robot



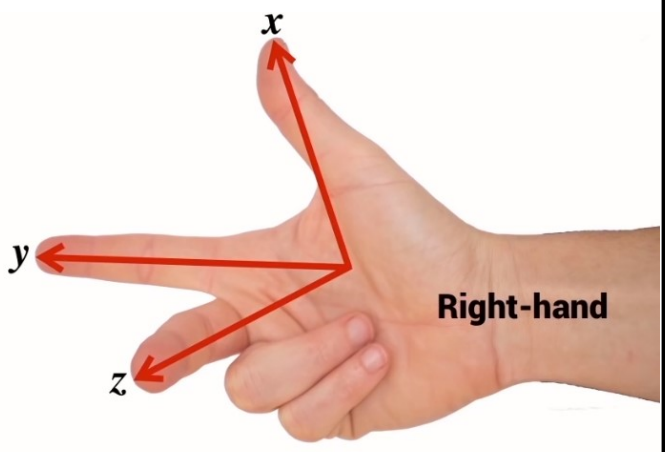
35

Example 6 – Six DOF Collaborative Robot

Joint i	θ_i (deg)	α_i (deg)	r_i (cm)	d_i (cm)
1	θ_1	90	0	a_1
2	θ_2	0	a_3	0
3	θ_3	0	a_5	0
4	θ_4	90	0	$-a_6$
5	$-\theta_5$	90	0	a_7
6	$-\theta_6$	0	0	a_8

36

Right Hand Rule



Another Right Hand Rule

- For angle measurement
- Roll your finger around the axis of rotation and your thumb should point towards the direction of axis of rotation, then your fingers will curl in the direction of positive rotation



Link Transformations for FK

42

Link Transformations

$$^{j-1}\xi_j \sim \mathbf{A} = \mathbf{R}_z(\theta_j)\mathbf{T}_z(d_j)\mathbf{T}_x(a_j)\mathbf{R}_x(\alpha_j)$$

$$^{j-1}\mathbf{A}_j = \begin{pmatrix} \cos\theta_j & -\sin\theta_j\cos\alpha_j & \sin\theta_j\sin\alpha_j & a_j\cos\theta_j \\ \sin\theta_j & \cos\theta_j\cos\alpha_j & -\cos\theta_j\sin\alpha_j & \alpha_j\sin\theta_j \\ 0 & \sin\alpha_j & \cos\alpha_j & d_j \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

// this is not alpha j.. this is a j.

43

Home Task

$$^{j-1}\xi_j \sim \mathbf{A} = \mathbf{R}_z(\theta_j)\mathbf{T}_z(d_j)\mathbf{T}_x(a_j)\mathbf{R}_x(\alpha_j)$$

Perform the matrix multiplications above to prove that A is equal to the following matrix.

$$^{j-1}\mathbf{A}_j = \begin{pmatrix} \cos\theta_j & -\sin\theta_j\cos\alpha_j & \sin\theta_j\sin\alpha_j & a_j\cos\theta_j \\ \sin\theta_j & \cos\theta_j\cos\alpha_j & -\cos\theta_j\sin\alpha_j & \alpha_j\sin\theta_j \\ 0 & \sin\alpha_j & \cos\alpha_j & d_j \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

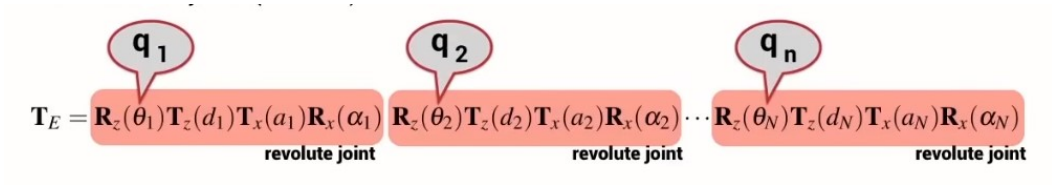
Link Transformations

- The parameters αj and aj are always constant.
- For a revolute joint, θ j is the joint variable and dj is constant, while for a prismatic joint, dj is variable, θj is constant and αj = 0.
- In many of the formulations that follow, we use generalized joint coordinates qj

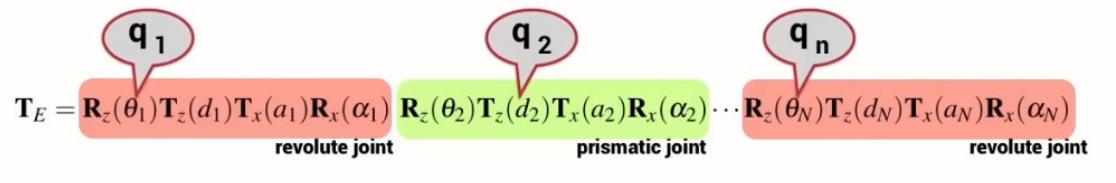
$$\text{if } \sigma_j = \begin{cases} R: & \theta_j \leftarrow q_j \\ P: & d_j \leftarrow q_j \end{cases}$$

For N link robot – Stack Poses

- All Revolute (RRRRRR)



- Revolute and Prismatic (RPRRRR)



Link Transformation

- To find the single transformation that relates frame [N} to frame {0}:

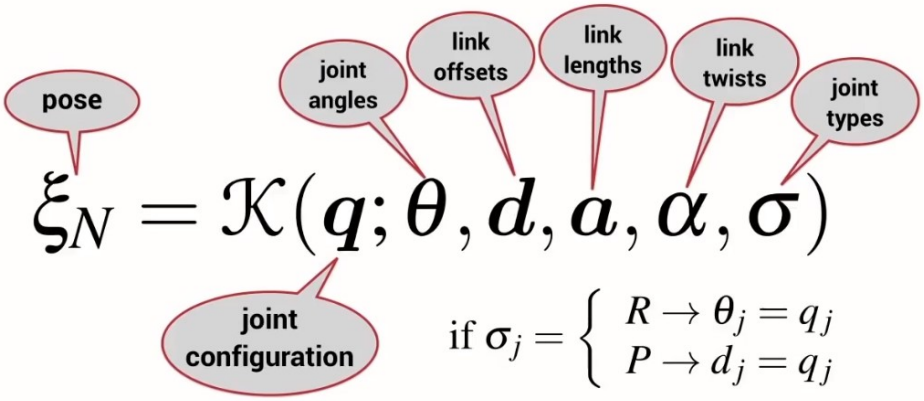
$$ {}^{0}_{N}T = {}^{0}_{1}T\ {}^{1}_{2}T\ {}^{2}_{3}T \ldots {}^{N-1}_{N}T $$

Forward Kinematics (K)

- The forward kinematics is a function of the joint coordinates and is simply the composition of the relative pose due to each link

$$ {}^{0}\xi_N = \mathcal{K}(\boldsymbol{q};\theta,\boldsymbol{d},\boldsymbol{a},\alpha,\sigma) = {}^{0}\xi_1 \oplus {}^{1}\xi_2 \cdots {}^{N-1}\xi_N $$

General Form



Activity Examples for Links Transformations

-



Example 1

- Compute the individual transformations for each link.

Link	a_i	α_i	d_i	θ_i
1	0	0	d_1	θ_1
2	0	-90	d_2	0
3	0	0	d_3	0

$$ {}^{j-1}\boldsymbol{A}_j = \begin{pmatrix} \cos\theta_j & -\sin\theta_j\cos\alpha_j & \sin\theta_j\sin\alpha_j & a_j\cos\theta_j \\ \sin\theta_j & \cos\theta_j\cos\alpha_j & -\cos\theta_j\sin\alpha_j & \alpha_j\sin\theta_j \\ 0 & \sin\alpha_j & \cos\alpha_j & d_j \\ 0 & 0 & 0 & 1 \end{pmatrix} $$

Example 1

$$^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^1T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 1

$$^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^1T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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53

Example 1

- Find the ${}^0T_3$

$$ {}^0T_3 = \begin{bmatrix} c_1 & 0 & -s_1 & -s_1 d_3 \\ s_1 & 0 & c_1 & c_1 d_3 \\ 0 & -1 & 0 & d_1 + d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Example 2

- Find the all the Transformation matrices

	θ	α	r	d
1	θ_1	90	a_2	a_1
2	90+θ_2	90	0	0
3	90	0	0	$a_3 + a_4 + d_3$

Example 2

- Find the ${}^0T_1$

$$\begin{bmatrix} C(\theta_1) & -S(\theta_1)C(90) & S(\theta_1)S(90) & a_2C(\theta_1) \\ S(\theta_1) & C(\theta_1)C(90) & -C(\theta_1)S(90) & a_2S(\theta_1) \\ 0 & S(90) & C(90) & a_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 2

- Find the ${}^1T_2$

$$\begin{bmatrix} C(90+\theta_2) & -S(90+\theta_2)C(90) & S(90+\theta_2)S(90) & 0 \\ S(90+\theta_2) & C(90+\theta_2)C(90) & -C(90+\theta_2)S(90) & 0 \\ 0 & S(90) & C(90) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 2

- Find the ${}^{2}T_{3}$

$$\begin{bmatrix} C(90) & -S(90)C(0) & S(90)S(0) & 0 \\ S(90) & C(90)C(0) & -C(90)S(0) & 0 \\ 0 & S(0) & C(0) & a_3 + a_4 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 2

- Find the ${}^{0}T_{3}$

$$\begin{bmatrix} C(90) & -S(90)C(0) & S(90)S(0) & 0 \\ S(90) & C(90)C(0) & -C(90)S(0) & 0 \\ 0 & S(0) & C(0) & a_3 + a_4 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 3

Joint number	θ_i	d_i	α_i	a_i
1	θ_1	0	0	l_1
2	θ_2	0	0	l_2
3	θ_3	0	0	l_3

$$T_i^{i-1} = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

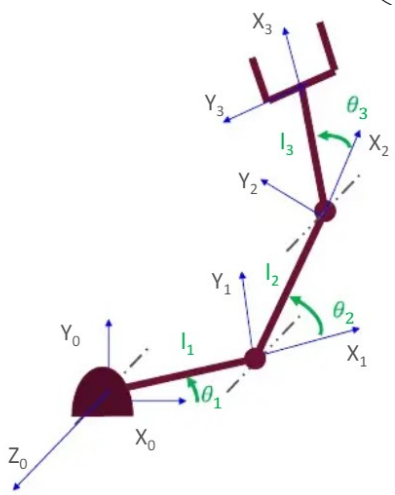
$$T_1^0 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & l_1 c\theta_1 \\ s\theta_1 & c\theta_1 & 0 & l_1 s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, T_2^1 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_2 c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & l_2 s\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, T_3^2 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & l_3 c\theta_3 \\ s\theta_3 & c\theta_3 & 0 & l_3 s\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

61

Example 3

Joint number	θ_i	d_i	α_i	a_i
1	θ_1	0	0	l_1
2	θ_2	0	0	l_2
3	θ_3	0	0	l_3

$$T_3^0 = T_1^0 T_2^1 T_3^2 = \begin{bmatrix} c\theta_{123} & -s\theta_{12} & 0 & l_1 c\theta_1 + l_2 c\theta_{12} + l_3 c\theta_{12} \\ s\theta_{12} & c\theta_{123} & 0 & l_1 s\theta_1 + l_2 s\theta_{12} + l_3 s\theta_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



62

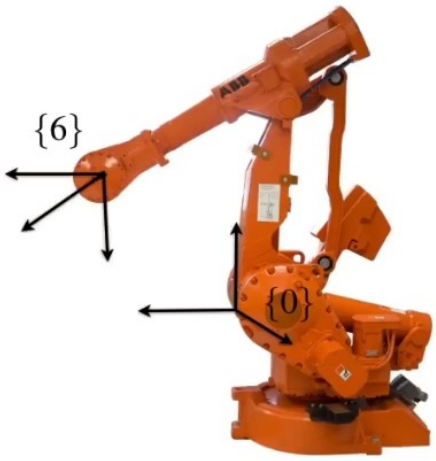
Base and Tool transforms



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82

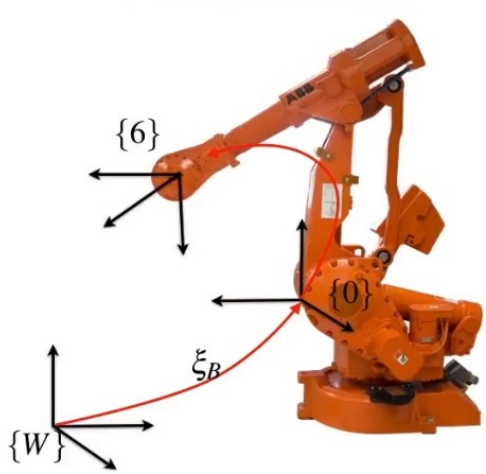
Base and tool transforms



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83

Base transform



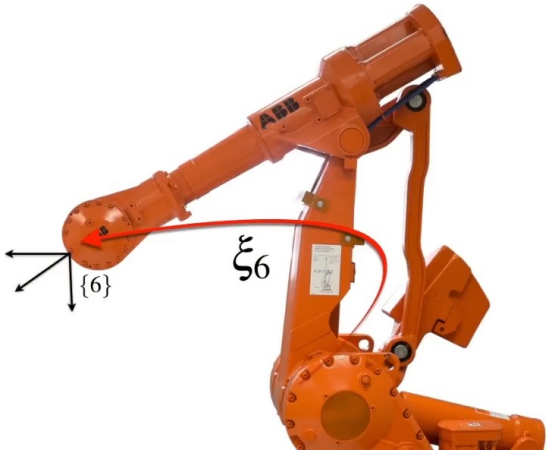


$${}^{0}\xi_6 = \mathcal{K}(q_1 \cdots q_6)$$

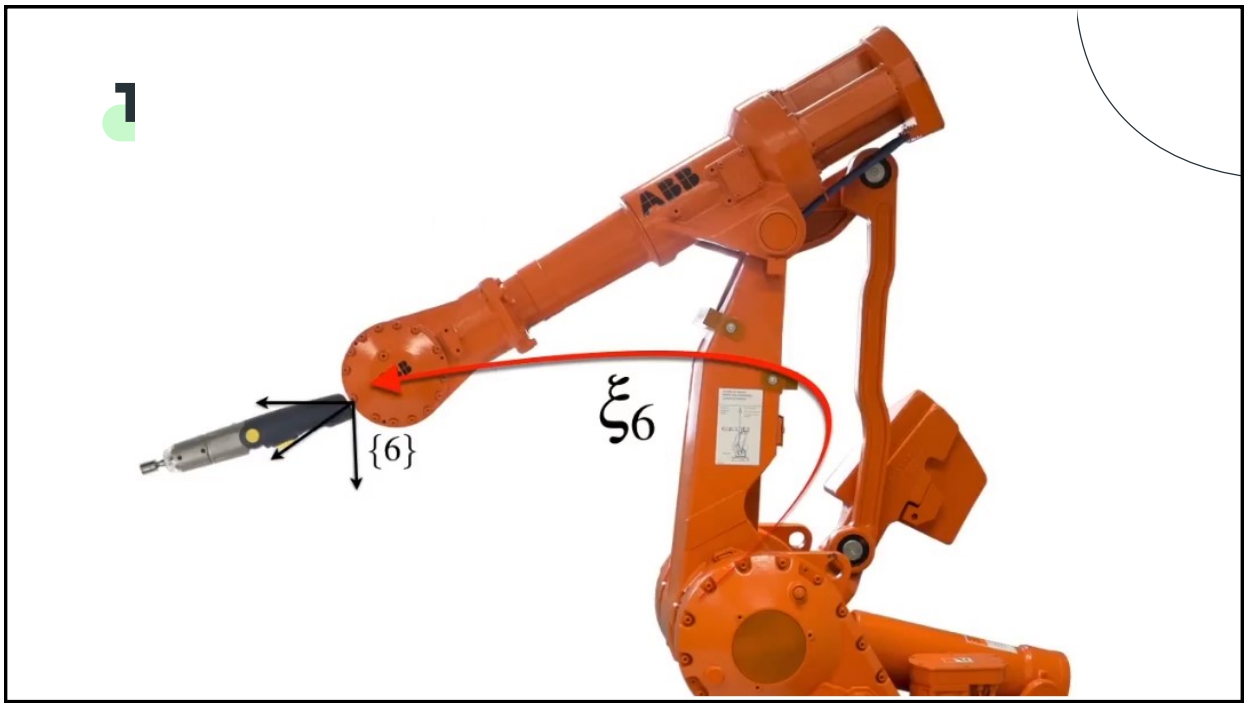
$${}^{W}\xi_6 = \xi_B \oplus \mathcal{K}(q_1 \cdots q_6)$$

84

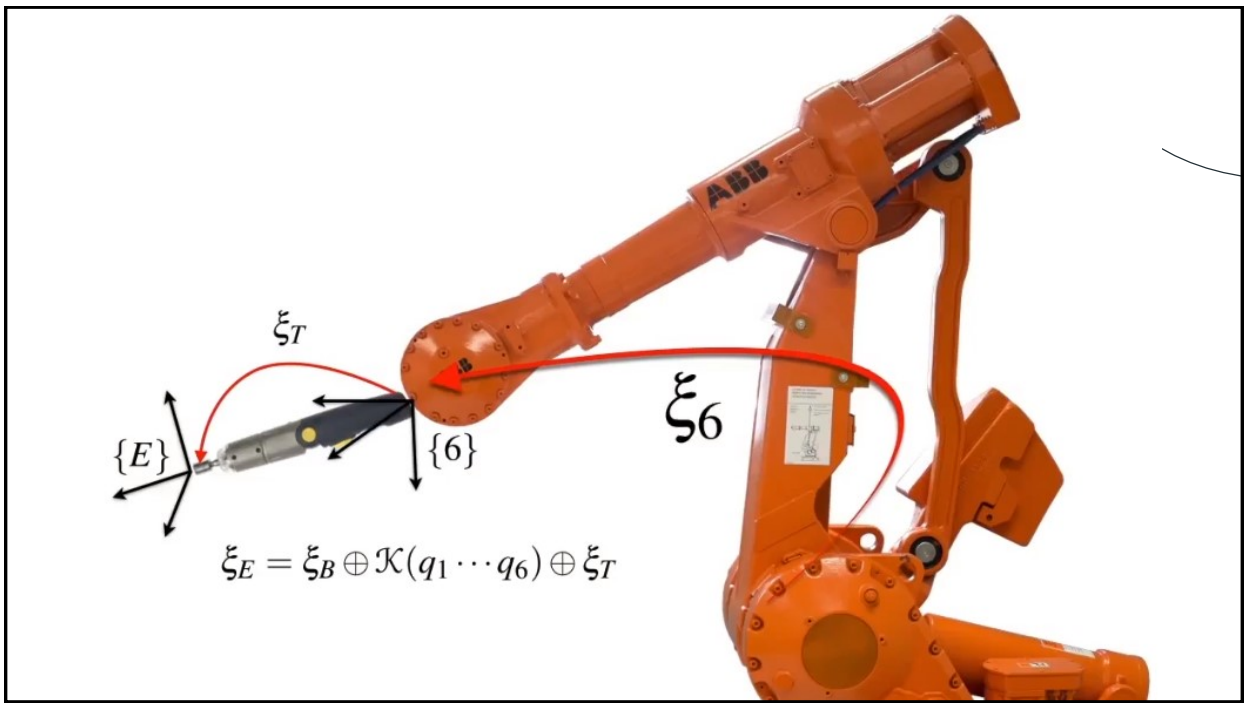
Tool transform



85


ABB
ξ_6
{6}

86


ABB
ξ_T
ξ_6
{E}
{6}
$\xi_E = \xi_B \oplus \mathcal{K}(q_1 \cdots q_6) \oplus \xi_T$

87

Robot workspace



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88

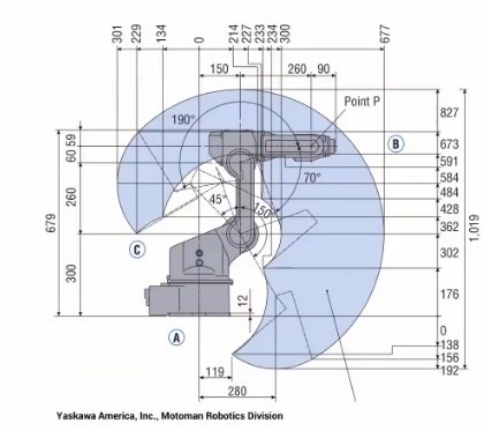
Robot workspace

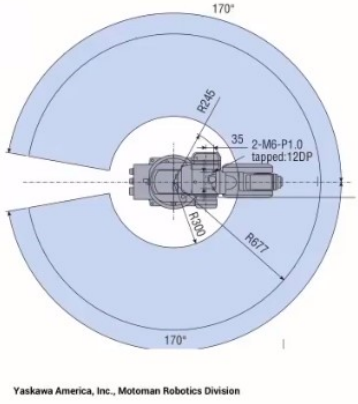
- The set of all reachable end point positions

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89

Robot workspace





Yaskawa Motoman SV3X

90



Let's Formalize some stuff

91

What is kinematics?



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92

What is kinematics?

- **Kinematics** is the branch of mechanics that studies the motion of a body or a system of bodies
- Concerned with position (and angles) and velocities (translational and angular)
- Not concerned with force or moments. Such a study is a part of Dynamics.

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93

Robot Kinematics

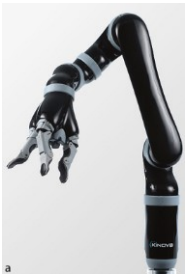
- Two kinematic problems in Robotics
- **Forward Kinematics**: Given joint angles, we can find the robot's tool tip.
- **Inverse Kinematics**: Given the pose of robot tool tip, we find the required joint angles.

Robot Arm

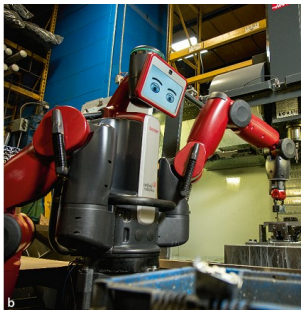
- A serial-link manipulator, comprises a chain of rigid links and joints.
- Each joint has one degree of freedom, either translational (a sliding or prismatic joint) or rotational (a revolute joint)

Robot Arm

- One end of the chain, the base, is generally fixed and the other end is free to move in space and holds the tool or end-effector that does the useful work.



***End-Effector is a complex function of the state of each joint**



96

Goals

- Forward Kinematics: How to compute the pose of the end-effector?
- Inverse Kinematics: How to compute the position of each joint given the end-effector pose?
- How to determine DH parameters?

97

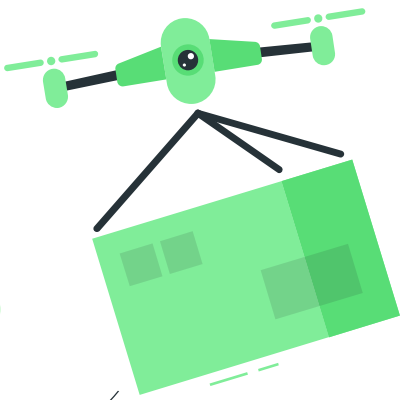
Forward Kinematics

- **Definition**: the mapping from joint coordinates, or robot configuration, to end-effector pose.

```
>> E.fkine( 30, 'deg')
ans =
    0.8660   -0.5000      0.866
    0.5000    0.8660        0.5
         0         0          1
```


98

Assignment 1



100

Assignment 1 – More Discussion

- **Goal:** Design a robotic manipulator.
- **Workspace**: Half doughnut.
- It is not hollow.
- Deliverables:
- Evaluation:



101

Forward Kinematics vs. Inverse Kinematics:
Forward Kinematics (FK): Given joint angles, determine the position and orientation of the end-effector.
Inverse Kinematics (IK): Given the position and orientation of the end-effector, determine the joint angles.

Inverse kinematics allows us to find the necessary joint configurations to achieve a desired pose of the robot's end-effector. It is a critical process in controlling robotic systems and can be solved analytically for simple cases or numerically for more complex robots.